Expt No: 3
Date: 23/7/26

UNDERSTANDING CISCO PACKET TRACKER

Aim: To understand the Cisco packet Tracker environment and begin a design a Simple Computer network using its interface and networking components.

Steps:

* Open the Cisco packet tracker application.
* Observe the UI including Menu Bar, Main tool bar, workspace and other.
* From the network component box, Select the required network device Such as PC's, HUB and Switch.
* Drag and drop the devices into the workspace.
* Connect the devices ~~into the~~ using Copper Straight through Cables.
* Configure IP address for the PC using the Desktop → IP Configuration option.
* Save the network topology.

Result: The cisco packet tracker environment was Successfully explored and a Simple network was designed. The interface Components, device Connections, IP Configuration and packet Tracker were Successfully understood using both realtime and Stimulation modules.

Exp No: 3.1

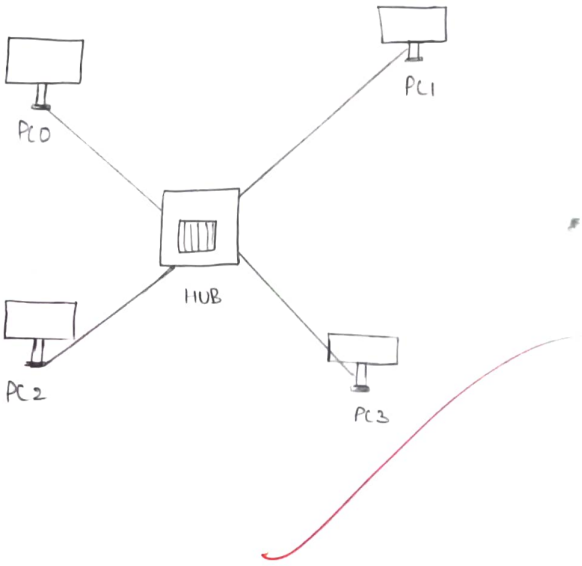
Date: 23/7/26

HUB

Aim: To Study the behaviour of a HUB by observing how its forwards data packets using cisco packet Tracer.

Steps:

* Open Cisco packet Tracer.
* Drag and place 1 Hub and 4 Generic PC's into the workspace.
* Connect the all PC's to the Hub using Copper Straight through Cables
* Configure IP addresses for all PC's in the same network.
* Select Add Simple PDU from the Common Tool Bar.
* Send a packet from one PC (Source) to another PC (destination).
* Switch to Simulation mode and observe how the HUB forwards the packets.



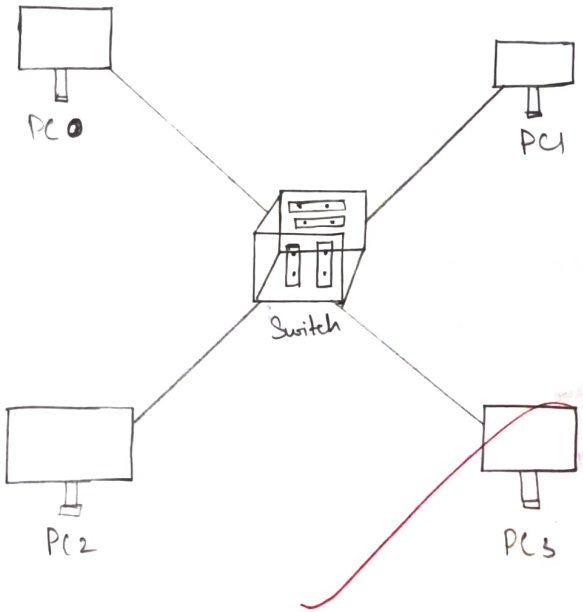
Result:

The HUB broadcasts the received packet to all connected devices. Every PC receives the packet but only destination PC accepts it, while the others discard it.

Expt No: 3.2

Date: 23/7/26

SWITCH



Aim: To Study The behaviour of a Switch by observing how it forwards data packets using Cisco packet tracker.

Steps:

* Open Cisco packet Tracker.
* Drag and place 1 Switch and 4 Generic PC into workspace.
* Configure IP addresses for all PC's.
* Connect all PC's to the Switch using Copper straight Through Cables.
* Select add Simple PDU from Common tool bar.
* Send a packet from one PC (Source) to another PC (destinati
* Switch to Simulation mode and observe how the Switch forwards the packet.

Result: The Switch forwards the packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves Communication efficiency.

Expt No: 3.3

Date: 23/7/25

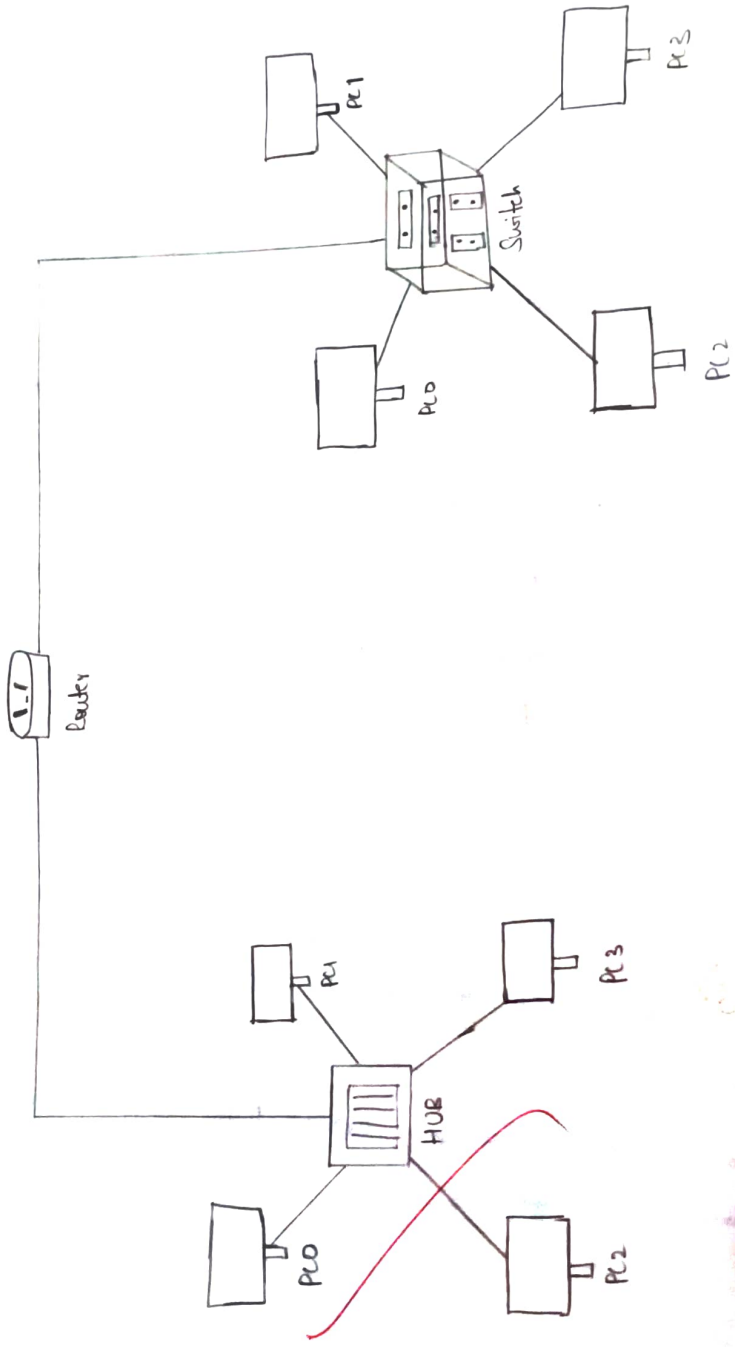
HUB And SWITCH (comparison)

Aim:

To Compare the packet forwarding behaviour of a Hub and a Switch using Cisco packet tracer.

Steps:

* Create one network using a Hub and another using a Switch, each Connected to 4 PC's.
* Configure IP addresses for all PC's in both networks.
* Send a Simple PDU between two ~~PC~~'s PC's in each network.
* Observe packet transmission in Simulation mode.
* Compare how the HUB and Switch forward packets.



Result: The HUB broadcasts packets to all Connected devices, Whereas the Switch forwards Packets only to the intended destination. Therefore Switch provides better performance, reduced network traffic and Improve efficiency Compared to the HUB.